Spatial Model Update

The Allocation of Teaching Talent and Human Capital Accumulation

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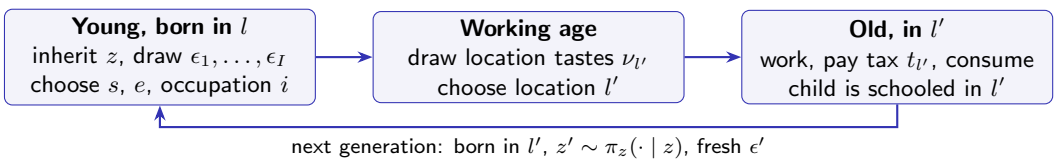
The spatial model overview

Kept from the aspatial model

- OLG with time (s) and goods (e) investment
- Roy choice over 21 occupations
- $h = h_T^{\beta} a^{\alpha} s^{\phi} e^{\eta} N (h_T)^{-\sigma}$, efficient class sizes
- Teacher pay κh^{γ}, $\gamma < 1$; gender wedges

New

- Ability $a_i = z\,\epsilon_i$, with z inherited
- Two locations with local teacher markets and budgets
- Location choice with a moving cost
- Altruism: parents choose their children's schools



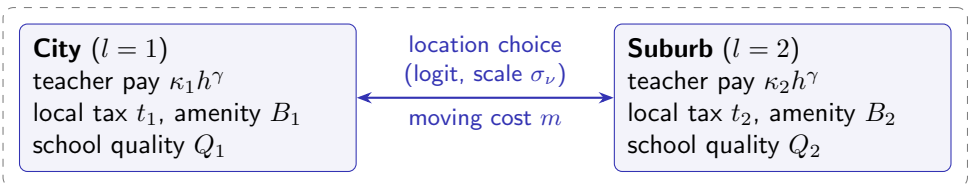
Ability: inherited z, occupation-specific ϵ

$$a_i = z\,\epsilon_i, \qquad \log z' = \rho_z \log z + \sigma_\xi\,\xi, \qquad \log \epsilon_i \overset{\text{iid}}{\sim} N\big(-\tfrac{1}{2}\sigma_\epsilon^2,\ \sigma_\epsilon^2\big)$$

- z is inherited $\Rightarrow$ families can sort across locations on z; ϵ_i is redrawn $\Rightarrow$ Roy selection
- Teaching pays κh^γ, $\gamma < 1$: z does not cancel $\Rightarrow$ high-z agents select out of teaching
- Elsewhere z and location cancel, and test scores load on z only $\Rightarrow$ estimate outside equilibrium:

Parameter	Target	Data	Estimate
Occupational dispersion σ_ϵ	Within-occupation wage 90/10 (ACS)	3.74	0.786
Persistence ρ_z	Mother–child latent score corr. (CNLSY)	0.577	0.577
Dispersion of z, s_z	Log wage per latent AFQT SD (NLSY79)	0.213	0.196

Two locations with local teacher markets



- School quality $Q_l = \left(2\widetilde{H}_{T,l}/M_l\right)^\sigma$: local teacher human capital per pupil
- Teachers are paid from a local balanced budget; class size is the only congestion force
- Baseline: live = work = school. Equilibrium is a fixed point in $(\widetilde{H}_{T,l}, M_l, t_l)$ with a stationary $\Phi_l(z)$

Altruism and the sorting problem

$$U = \ln(1-s) + \mathbb{E}\big[\,\mu \ln C + B_{l'} + \sigma_\nu \nu_{l'} + \lambda f(h')\,\big], \qquad h' = \text{child's human capital}$$

- Parents do not fund the child's investment, so location l' is worth

$$V_{l'} = \mu \ln C_{l'} + \Lambda_{l'}(z), \qquad \Lambda_{l'}(z) = \lambda\, \mathbb{E}\big[f(h') \mid z,\, l'\big]$$

  $\Rightarrow$ altruism acts only through location choice
- **Neutrality:** with $f = \log$ and a utility moving cost, location premia are proportional
  $\Rightarrow$ no sorting on ability
- Two departures, no new free parameters:
  1. **Goods moving cost** $m = r_m \bar{C}$: weighs more on the poor
  2. **Linear warm glow** $f(h') = h'/\bar{h} - 1$: schools matter more for abler children

Data: city vs. suburb within a commuting zone

- City (NCES 11–13) vs. suburban (21–23) districts, 188 CZs
- Moments: within-CZ gaps, enrollment-weighted

Schools look alike; families do not ⇒ the gap is mostly who lives there

Suburb − city	Gap (SE)
Schools	
Teacher salary, log†	0.009 (0.009)
Teachers per pupil, log	−0.018 (0.013)
SEDA learning rate, SD/grade	0.003 (0.003)
Families	
Median FTYR earnings, log	0.112 (0.017)
Child poverty, pp	−8.9 (0.9)
SEDA score level, SD	0.289 (0.033)
Enrollment, log†	0.376 (0.115)
City↔suburb move rate†	0.216 (0.011)

† targeted in the calibration

Sources: CCD, F-33, CWIFT, SAIPE, ACS (districts, 2018); SEDA 2009–19; NLSY79 (move rate).

Calibration: eight parameters in equilibrium

Fixed: $\sigma = 0.56$ (class size, FOO), $\sigma_\nu = 0.25$ (Eckert–Kleineberg), $\chi = \log 1.12$ (CFR). First stage: A_i and female wedges match occupation shares by gender.

Parameter	Value	Target	Source	Data
Pay scale $\log \tilde{\kappa}$	-1.204	Male teaching share	ACS	1.95%
Female outside option ω_f	0.966	Female teaching share	ACS	5.99%
Teacher wage curvature γ	0.619	Teachers' wage 90/10	ACS	2.77
Suburb pay premium δ_κ	0.0045	Salary gap	F-33	0.0086
Suburb amenity ΔB	0.066	Enrollment gap	CCD	0.376
Teacher spillover β	0.228	Earnings per SD-year of teacher quality	CFR	1.34%
Altruism λ	0.125	WTP for school scores, % of C	BFM	0.60%
Moving cost r_m	0.206	City↔suburb move rate	NLSY79	0.216

- Exactly identified: all eight matched; 15 dispersed starts find no other root
- Jacobian full rank (condition number 30); each spatial moment loads on its own parameter

Validation: the fit misses sorting and teacher allocation

Untargeted moment	Data	Model
Household sorting		
Earnings gap, log	0.112	−0.010
SEDA score gap, SD	0.289	0.061
Child-poverty gap, pp	−8.9	+0.2
City share, top − bottom tercile	−0.295	+0.011
Teacher allocation		
Teachers-per-pupil gap, log	−0.018	+0.023
Learning-rate gap, SD/grade	0.003	0.012
Teacher selection		
Teacher / non-teacher schooling	1.18	0.76
Teachers' ability vs. others, SD	+0.61	−0.29
Teacher pay vs. market, m/f	+.06/+.09	−.72/−.63

- **No sorting:** the suburb's school edge is worth ≤ 0.002 utils vs. $\sigma_\nu = 0.25$; WTP pins λ, SEDA growth keeps $\Delta \log Q$ small
- **Over-staffing:** matching the small pay gap over-staffs the suburb and inflates its learning-rate edge
- **Negative selection:** with $\gamma < 1$, high-z agents avoid teaching; short schooling fills it at low pay